Predicting the Second Gift

Which first-time donors will give again, and which ones a small nonprofit should spend its scarce staff hours on.

Team Bakul Badwal · Malorie Black · Reid Jacobson · Thadeus Knospe · Rodolfo Perez-Cortes Manrique

STAKEHOLDER	DECISION	DATA	GROUND TRUTH	HONEST BASELINE
Development lead at a ~$500K nonprofit, no data staff	Which first-time donors get a personal follow-up, and where the cutoff sits	DonorsChoose Open Data 2002–2019 (ICPSR 37898): 11.4M donations, 2.1M projects	Observed second gift within 12 months, held-out 2017–18 cohorts	RFM — the recency-and-amount rule small nonprofits actually use

01 PROBLEM AND BUSINESS APPLICATION

Most nonprofits acquire a donor once and never hear from them again. The second gift is the highest-leverage moment in the donor lifecycle, and small organizations are the least equipped to work it. Our stakeholder is the development lead at a nonprofit with roughly a $500K budget and no data staff: each month she can personally follow up with only a fraction of first-time donors, and today she builds that list by hand from recency and gift size.

The decision is specific: given a fixed outreach budget in staff hours, which first-time donors receive follow-up, and where does the cutoff sit? Our output is a ranked follow-up list with a recommended threshold and the expected dollars behind it. One of us founded GoGood Technologies, a donor-engagement platform serving exactly this customer, so real practitioners can check the output. **In practice:** the hand-built list becomes a scored one, and the budget question gets a number.

02 DATASET AND SOURCE

DonorsChoose Open Data, United States, 2002–2019 (ICPSR 37898). Public-use files, free with a no-cost ICPSR account. Two files: Donations (11,377,479 records) and Projects (2,149,817), covering September 2002 to June 2019 with activity through December 2019. Donations carries DONOR_ID, so we reconstruct each donor's history and build our own label, plus amount, month, donor type, and matched, teacher-referred, and thank-you-packet flags. Two constraints accepted up front: dates are month-level, and there is no demographics file, so donor type is the only donor attribute.

03 WHAT WE WILL BUILD

04 EVALUATION — WHAT WE MEASURE, AGAINST WHAT GROUND TRUTH

Ground truth is observed donor behavior in the held-out cohorts, not hand labels. Four measurements:

- **Discrimination and calibration.** PR-AUC because the class is imbalanced, plus a calibration curve: an uncalibrated score cannot support a threshold.
- **Honest baselines.** RFM, the recency-and-amount heuristic small nonprofits actually use, plus random targeting and contact-everyone. Business metric: net value retained per hour of outreach at a fixed budget. If the model does not beat RFM, that is the finding.
- **Does stewardship predict retention?** The thank-you-packet flag records a real post-gift action. We estimate its association with a second gift controlling for gift size; packets were not randomly assigned, so association, not cause.
- **Error analysis by cohort.** We expect the model to be weakest on donors with the thinnest history, the case the stakeholder cares about most.

> **Limitation we test rather than assume.** DonorsChoose donors are marketplace donors; small-nonprofit donors are relational. We test which signal survives: behavioral features versus platform-specific ones.

05 WHAT WE NEED FROM YOU

- Confirmation that a well-supported negative finding against the RFM baseline is an acceptable result.
- Guidance on ~13 GB on JupyterHub, or approval to use a stratified sample of donor cohorts.
- Any concern about anchoring the framing to a company a